

VENDX.

YOUR SENSORS. THEIR WALLETS.

A five-dollar sensor that sells its own readings — and gets *paid* for them, in seconds, by software that no human had to introduce it to.

What it is, how the money moves, and who would actually buy the data — in plain language, assuming no background in hardware or payments.

INSIDE

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- 02 How the money moves
- 03 What keeps it honest
- 04 A use case, end to end
- 05 Where else this fits
- 06 What actually runs today
- 07 The words, in plain English

\$0.01

the price of one reading

~40 ms

to check that payment

378 bytes

the whole proof, on the wire

A vending machine, but the product is information.

Put a coin in a vending machine and a can comes out — no contract, no invoice, and the machine never learns who you are. VENDX gives that same arrangement to small sensors.

THE WHOLE SYSTEM, IN ONE SENTENCE

A cheap sensor is asked for its data, replies “that costs one cent”, gets paid, checks the payment itself in about the time it takes to blink, and hands the data over. No account, no invoice, nobody in the loop.

HOW IT WORKS TODAY

Data is sold in bulk, slowly.

Somebody negotiates a contract and agrees a monthly fee. It takes weeks.

Small sensors can't take part.

One air-quality monitor cannot invoice anyone, so its readings are thrown out.

Software buyers are second-class.

An assistant sent to gather data cannot pay for it. It scrapes, then guesses.

HOW VENDX WORKS

Data is sold by the reading.

One request, one cent, settled in seconds. No paperwork to amortise.

The sensor is the merchant.

It sets its own price, holds its own wallet and checks its own payments.

Software can pay, within limits.

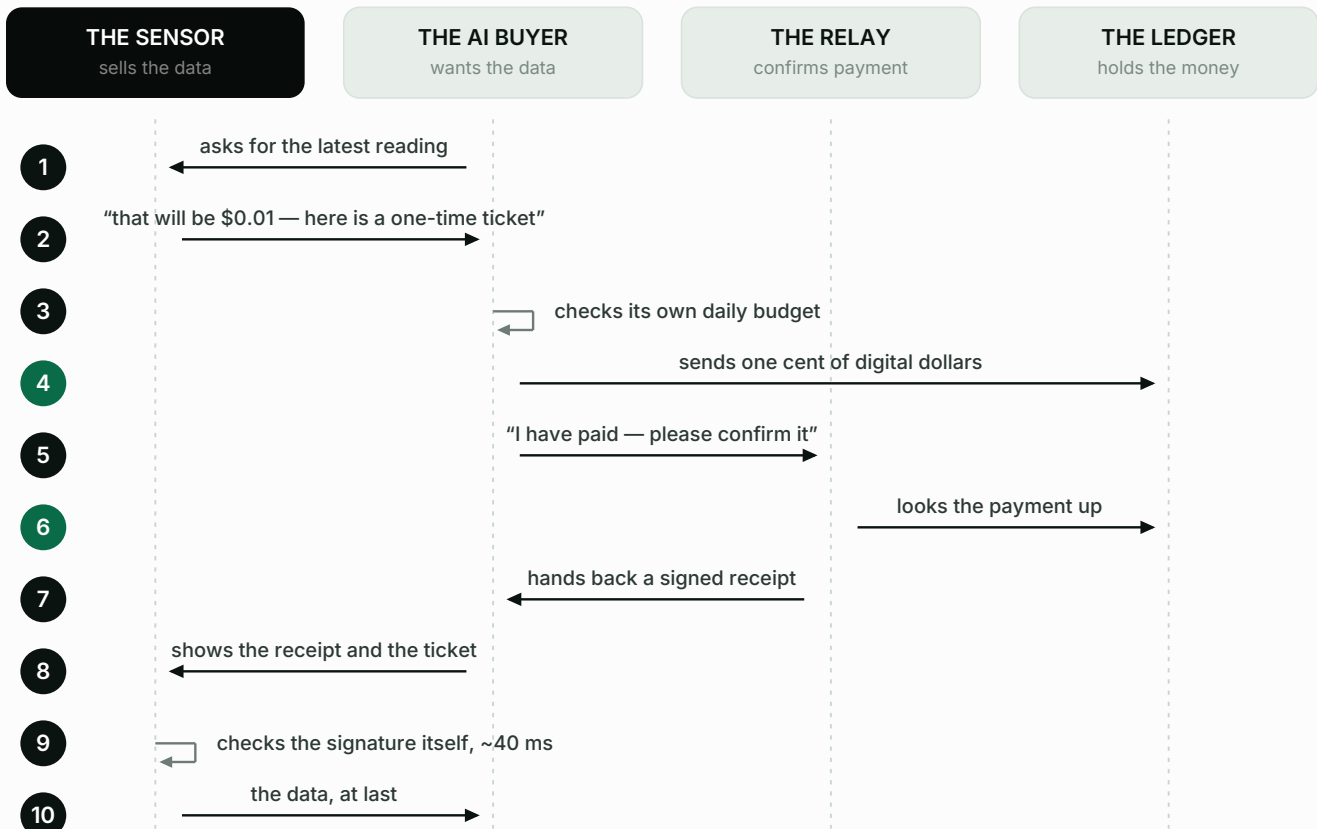
The buyer carries a budget — five dollars a day — and is refused past it.

WHY NOW

Two things changed at once. Software now runs real errands and hits paywalls with no way through, and a cent of digital money moves in about a second. VENDX points those two facts at each other.

Ten steps, and nobody is watching.

This is the whole transaction. It runs with no person present, usually in two or three seconds.



Steps 4 and 6 are the only two that touch a public ledger — the shared record proving the cent really moved. The rest is ordinary web traffic.

THE ONE PIECE THAT IS NOT OBVIOUS

Why is there a middleman at all? Because the sensor is too small to read the ledger itself — opening a secure connection would need more memory than it has. So the relay looks on its behalf and writes it a receipt. The next page is why that receipt is hard to fake.

A five-dollar chip checking money, like a shopkeeper checking a note.

The sensor cannot phone anyone to ask whether a payment was real. So it checks the receipt the way you would check a banknote.

✓ **Is the signature genuine?**

It holds one key, fixed when it was made.

✓ **Is this ticket mine, and fresh?**

Unknown or expired tickets are refused.

✓ **Has this ticket been spent?**

A receipt works exactly once.

✓ **Was I the one who got paid?**

A receipt made out elsewhere is worthless.

✓ **Was it enough money?**

Paying less than the price is not paying.

✓ **Right currency, right network?**

A practice receipt is not spendable for real.

AND THE PART WE ARE NOT PROUD OF

The sensor has to trust one party.

Because the chip cannot read the ledger itself, it trusts the relay's word that a payment happened, and recognises its signature. A dishonest relay could vouch for a payment nobody made — a real limit, stated here rather than buried. What it cannot do is redirect, discount or re-use a genuine payment: the sensor checks all of that itself.

ON PRIVACY — THE PART THAT MATTERS TO PEOPLE

What gets sold is a **count**, never an identity. A phone walking past is reduced to a one-way fingerprint before it is counted, so the data can say “eleven people went by” and never who. The badge in this build also holds its owner's name, email and a wallet key in plain text — none of it sold, logged or published. If you carry one: treat its wallet as disposable.

Who would actually buy this? Start with a shopping mall.

A leasing manager wants to know whether the quiet east corridor is really quiet, or only quiet on Tuesdays. Today she pays a survey firm for a month-long count.

THE SELLER

34 door sensors already in the mall, counting passing phones.

THE BUYER

An assistant with a budget and one question to answer.

THE TRADE

One cent per five-minute count, paid on the spot, no contract.

WHAT HAPPENS, HOUR BY HOUR

- 09:00 ● **The question is asked.**
“How busy is the east corridor, in five-minute slices, over the last fortnight?”
It is given five dollars a day and sent off.
- 09:00 ● **It finds sellers and gets quoted.**
It asks each sensor directly and is quoted one cent a reading, with a one-time ticket. No sign-up, no paperwork, no sales call.
- 09:01 ● **It does the arithmetic itself.**
A fortnight of five-minute readings is 4,032 per sensor — about forty dollars. So it narrows the order to three sensors, opening hours only.
- 09:03 ● **It buys, and the money lands.**
Each purchase is one cent, settled in about a second. Each sensor checks its own payment and releases its own reading.
- 09:40 ● **The answer arrives — with a surprise.**
The corridor is busy from 11:40 to 13:20 and dead the rest of the day. That is a lunch-trade corridor, not a dead one.
- tomorrow ● **And it can simply be asked again.**
The same question next week costs the same \$2.16, with nothing to renegotiate. The corridor is now monitored rather than surveyed.

Nobody signed anything.

No contract, no onboarding, no invoice, no monthly minimum. The sensors earned money from a customer they had never heard of.

WHAT EACH SIDE GETS OUT OF IT

The buyer pays

\$2.16

216 readings at a cent each, over twelve hours

Each sensor earns

\$0.72

for data it was already collecting and threw away

The survey firm

not asked

quoted in weeks, not minutes

Illustrative, at the one-cent price used in the demo. The point is not the figure — it is that a seventy-two-cent sale is worth making at all, because nobody had to do paperwork to make it.

THE OBVIOUS OBJECTIONS

“A cent is too small to bother with.”

That is the point. What stops tiny sales today is not the amount, it is the overhead around it — the account, the contract, the invoice.

“What stops the buyer from simply not paying?”

Nothing needs to. The sensor hands over nothing until it has checked the payment. There is no debt to chase because there is never any credit.

“What stops a buyer draining its owner's wallet?”

A cap it cannot talk its way past — five dollars a day — with every refusal logged. The demo has a button that tries to overspend so you can watch it hold.

Anywhere one measurement is worth a cent to somebody.

01

Air quality on one street

A city has a hundred monitors and no budget to serve their data. A journalist's research agent buys the one block it cares about, for the week it cares about, and the monitors pay for their own upkeep.

02

Is the car park full?

A navigation app does not want a data partnership with every car park in the country. It wants one number, right now, from one entrance, for less than it costs to think about — and then never to think about it again.

03

Soil moisture on a rented field

A co-op's probes sell to the insurer, the agronomist and a weather model separately, each paying for what it reads — without the farmer brokering three separate contracts.

04

Freezer temperatures overnight

A supplier's compliance agent buys last night's log from the shop itself, instead of asking staff to email a spreadsheet that nobody ever fills in. The shop gets paid for the trouble.

05

Is the old bridge still fine?

Vibration sensors sell to the engineer who inspects it, the haulier routing lorries over it and the council that owns it — three buyers, one set of readings, no data partnership.

06

How loud was it last night?

A residents' association buys the hour it is arguing about, from the two streets it is arguing about, rather than commissioning a noise survey it cannot afford and waiting a month for it.

THE PATTERN

Every one of these is the same shape: a measurement that already exists, a buyer who wants a slice of it, and a deal too small to survive a contract. That deal is not lost because the data is missing. It is lost because there was no way to charge a cent for it.

What is real, and what merely compiles.

A demo that overstates itself is worthless the moment somebody looks. Here is the state of the build, with nothing rounded up.

✓ working, checked end to end

● written and builds, not running

✗ deliberately not attempted

✓ The payment handshake	Runs start to finish against the stand-in sensor. Receipts are signed, verified, and refused when tampered with.
✓ Readings from real hardware	A genuine Hack the North badge is attached and its true readings are what get sold. Nothing on it was reflashed or erased.
✓ The public website	Twelve pages, deployed. Every figure says where it came from, and anything simulated says so on screen.
✓ The buyer's spending limit	A five-dollar daily cap that survives a restart, with a log of every purchase it refused to make.
● The sensor's own program	Compiles cleanly but is not running on the badge: that firmware is locked by the organisers. A relay reads the badge over a cable instead.
● The cheap-history program	Builds, 197 KB, not deployed. It is what would keep years of readings affordable; nothing in the demo depends on it yet.
✗ Real money	Everything settles on a practice network with practice dollars. No real-money transaction has been made, or is claimed.
✗ Our own app on the badge	The badge refuses side-loaded apps — verified by experiment, not assumed. Ours waits for a route that does not defeat that guard.

Why say all this out loud? Because the interesting claim is not “we built a lot”. It is “a five-dollar sensor can check a payment by itself in forty milliseconds”. That is either true or it is not, and hiding the unfinished parts would make it impossible to tell.

Eight words, and then you can read anything else about this.

These are the working definitions; precise ones live elsewhere.

"402"

A web error code that has existed, almost unused, since the 1990s: *payment required*. VENDX is an answer to it.

The ledger

A shared public record of who paid whom, so nobody has to be taken at their word about whether a payment happened.

A ticket (a "nonce")

A one-time number inside every quote. It is what stops a receipt being used twice.

Compressed history

A way to keep years of readings provable without paying storage rent on every single one of them.

Digital dollars

Money that moves like a message. A cent can be sent in about a second, and sending it costs a fraction of a cent.

A receipt

A short signed note saying "this payment cleared". Small enough — 378 bytes — that a five-dollar chip can check it.

The relay

The middleman that watches the ledger for the sensor and signs the receipt. The one party the sensor must trust.

The chip

An ESP32-C3. One small processor, about 28 KB of memory free while its radio is on, roughly five dollars.

IF YOU WANT THE LONG VERSION

See it running

[web-rouge-six-46.vercel.app](#)

The live site: the sensors, the buyer's console, the settled payments.

How it is built

[docs/ARCHITECTURE.md](#)

Every moving part, what it owns, and where the seams are.

The exact wire format

[docs/PROTOCOL.md](#)

What is sent, byte for byte, and why the obvious library was not used.

The badge, measured

[docs/BADGE.md](#)

What the hardware really is, what it refuses to do, and how we found out.

IN ONE LINE

A sensor that can charge for its own data stops being *equipment* and starts being a business.

That is the whole bet. Every number in this brief — the one-cent price, the forty-millisecond check, the 378-byte receipt — exists to make a sale that small worth making.

SEE IT RUNNING
web-rouge-six-46.vercel.app

THE WIRE FORMAT
[docs/PROTOCOL.md](#)

HOW IT IS BUILT
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THE BADGE, MEASURED
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